

ASSIGNMENT - NETWORK FORENSICS WITH TCPDUMP & TSHARK

OVERVIEW

You recently received a Linux application (binary) that is suspected of being malware (do not worry, it is not). You have been assigned to capture all network traffic that is generated from executing the binary.

INSTRUCTIONS



Case Number: YY-T004

For this assignment, you will need your Greymhatter virtual machine. Start your Greymhatter machine, navigate to Canvas and download the **network-forensics-client-ARCH** to your Desktop. The ARCH is your system architecture: x86_64 (AMD64) or Arm64; this is typically **amd64**. Once the binary is downloaded, you will need to make it executable before running it. Perform this by opening a terminal window, navigate to the Desktop folder, and make the file executable using the **chmod +x** command.

```
cd ~/Desktop                # navigate to desktop
chmod +x network-forensics-client-ARCH # make the binary executable
```

Next, we will need to figure out which network interface to monitor. We perform this by running the following command and finding the entry with an IPv4 address. Typically this will be **ens160**.

```
ip a # list all network interfaces
```

BEFORE STARTING THIS PROCESS, CLOSE ALL WEB BROWSERS OR ANYTHING THAT CONNECTS TO THE INTERNET!!!!!! Once we know the interface to monitor, we will need to start the packet capture prior to executing the binary. Start the packet capture by running the following command:

```
cd ~/Desktop                # navigate to desktop
sudo tcpdump -i INTERFACE -w malware-capture.pcap # run tcpdump
```

Now that we have tcpdump monitoring, we need to execute the binary. Open a new terminal window, navigate to the binary's location (Desktop), and run it by typing **./network-forensics-client-ARCH**. Do not remove with **./** in the command. You should get the following output:

```
./network-forensics-client-ARCH
🐱 Am I malware?
✅ Internet access confirmed.
✅ Successfully reached API.
✅ Host information sent successfully.
❌ Failed to download payload.
```

Once the last line is done, go back to your tcpdump terminal window and press **Ctrl + c**. This will end the packet capture. From here you should have **malware-capture.pcap** on your Desktop. Using **tshark** answer the following questions regarding the packet capture you just took.

QUESTIONS

- 1. Looking at IP conversations, what is the destination IP address of the conversation with the highest number of frames in and out?
- 2. Looking at TCP conversations, what is the destination port number for the IP address in the first question?
- 3. Looking at UDP conversations, what protocol is associated with the port found in all conversations?
- 4. Provide a sorted list of source IP addresses, with their respective counts.
- 5. Provide a sorted list of destination IP addresses, with their respective counts.
- 6. Provide a sorted list of DNS queries, with their respective counts.
- 7. Looking at hosts, what are the IP addresses for the hostnames in the DNS queries from the last question?
- 8. Looking at the protocol hierarchy, what is the top UDP protocol in this capture?
- 9. Looking at the protocol hierarchy, what is the top TCP protocol in this capture?
- 10. Based on the information from the last question, what do you conclude with regards to all TCP communications?
- 11. Where there any http requests?
- 12. Where there any http requests by a server?
- 13. How many connections were present for ICMP?
- 14. How many connections were present for DNS?
- 15. How many connections were present for SMB?
- 16. How many connections were present for TLS?

EVIDENCE

Filename	network-forensics-client-amd64
Type	ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=6b930aba937ea8b266da307682598347a189c744, for GNU/Linux 3.2.0, not stripped
Size	3962512 bytes
MD5	821A97B0CCE1579BF3A01A7AFB261B78
SHA1	4DAB0CBED2253B4CCFB207DE1440CF0F423BBEB1
SHA256	BCF65521AD23F5FE3DAAC8AB5D8256D9E50F8B98AEA4F84A0E3223AE91646544
SSDeep	49152:QyJzRNbmt5fXyyBEikxvAMuU14NZ6V6loa51fA/V8BkKrCuzgnUyC4gZwJrHViJt:Qy9mszq51fAOTvJ4dZ56qqfi+F

Filename	network-forensics-client-arm64
Type	ELF 64-bit LSB pie executable, ARM aarch64, version 1 (SYSV), dynamically linked, interpreter /lib/ld-linux-aarch64.so.1, BuildID[sha1]=2c55c30acf98df761603f5235ee63448bebd941d, for GNU/Linux 3.7.0, not stripped
Size	4197216 bytes
MD5	72E5D2FDB2C31D3711ED0FE0788AEFBB
SHA1	CDD7CDF383325575ED3CB046987CF3ABAC1F22C0
SHA256	33B336278B816FD1DFCDE5217807A355E7A9FBD1652134194EBD17F58A413DAD
SSDeep	24576:l55+hyLaOkrgwOfFDoTMH82j4xSZ9etvVpwPHaWf+EuNcKTkbQ5f7QbNrlEduHtq:l55zLyqEa9uNzTVQbNIEUHtcYd2GriY

REQUIRED TOOLS

Name	Vendor
tcpdump	GNU
tshark	Wireshark
wireshark	Wireshark
Greyhatter	Stauffer

REQUIRED FILES

- evidence-report-template.docx

DELIVERABLE REQUIREMENTS

- Evidence report with the following sections and requirements
 - Executive Summary
 - Evidence Seized
 - Tools Used
 - Case Information, Observables, Investigator Processes
 - Question answers
 - Conclusion
 - Exhibits (Not required)
 - References (Not required)
 - Photos/screenshots, w/ captions, are required to justify your findings
 - In line with your question answers, not in an appendix/exhibit
 - Must use report template

GRADING RUBRIC

- Evidence report - 100 pts
 - Completed answers - 64 pts
 - Answered questions for each item - 2 pts
 - Provided screenshot for each item - 2 pts
 - Used template - 26 pts
 - Following instructions - 10 pts